



COMSATS University
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Assignment # 02

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Course: Business Process Engineering

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Business Process Engineering (BPE) Evaluation Report

Subject: Business Process Re-engineering & Optimization Analysis

Target Focus: Process Evolution (V1 to V3) of 5 Global Market Leaders

1. McDonald's (Food & Retail Giant)

Process 1: Kitchen Order Fulfillment (Speedee Service System)

- **V1 (The Baseline):** Purely assembly-line system driven by human coordination. Orders were shouted out or printed on paper slips; staff manually compiled burgers step-by-step.
- **V2 (First Optimization):** Introduction of **KDS (Kitchen Display Systems)**. Electronic screens replaced paper chits. Kitchen stations were split dynamically based on real-time orders to balance workload.
- **V3 (Current Intelligent Stage): Predictive AI-Driven Kitchens.** Inventory systems talk directly to predictive kitchen monitors. Food prep dynamically accelerates *before* the customer reaches the window by tracking incoming drive-thru traffic queues.

Process 2: Customer Ordering System

- **V1 (The Baseline):** Counter ordering only. Customers had to queue, speak to a cashier, and pay with cash/card manually.
- **V2 (First Optimization):** Self-Service Kiosks. Customers select items from a digital screen, shifting the labor data entry task from cashiers directly to customers.
- **V3 (Current Intelligent Stage):** Mobile App & Personalized Geofencing. The mobile app matches past order preferences using machine learning. As soon as the customer crosses a specific GPS radius near the physical location, the kitchen automatically switches state to cook the meal hot.

2. Amazon (E-Commerce & Logistics Giant)

Process 1: Warehouse Picking & Sorting Process

- **V1 (The Baseline):** Manual Warehousing. "Pickers" walked miles down aisles with a clipboard paper sheet, physically looking for items on shelves and putting them into shopping carts.
- **V2 (First Optimization): Kiva Robotics System.** Instead of human pickers walking to the shelves, automated ground shelves move physically over wheels and navigate directly to stationary workers.
- **V3 (Current Intelligent Stage):** Computer-Vision & Intelligent Stacking. AI algorithms group items together dynamically based on predictive purchasing trends. Automated robotic arms use smart sensors to sort thousands of unpredictable shapes per minute.

Process 2: Delivery & Shipping Optimization

- **V1 (The Baseline):** Standard third-party courier shipping (postal services). Packages were shipped as they piled up, with static delivery path charts.

- **V2 (First Optimization):** Proprietary Hub-and-Spoke fleet distribution. Amazon created its own internal sorting centers and used dynamic software to find the shortest road routes for delivery trucks.
- **V3 (Current Intelligent Stage): Anticipatory Shipping & AI Routing.** Amazon literally ships packages toward regional hubs *before* local customers even click "Buy," predicting demand via historical clickstreams.

3. Starbucks (Beverage Retail Giant)

Process 1: Customer Beverage Customization & Ordering

- **V1 (The Baseline):** Baristas manually wrote custom order shorthand with sharpie markers directly on paper cups. Communication gaps often caused incorrect orders.
- **V2 (First Optimization):** Digital POS (Point of Sale) thermal printers. Cashiers punch orders into computers, which print out neat stickers with clear modifiers to stick on cups.
- **V3 (Current Intelligent Stage): Deep Brew AI Core.** The mobile app handles more than 30% of global queues. The backend engine coordinates inventory levels, barometric pressure (affecting espresso machines), and staff capacity to change digital menu boards instantly.

Process 2: Inventory & Supply Chain Management

- **V1 (The Baseline):** Manual ledger counts. Store managers walked around stock rooms at closing time checking boxes to fill out supplier sheets.
- **V2 (First Optimization):** Barcode scanning ERP software. Semi-automated inventory cycles where weekly scans flagged automatic item re-orders once thresholds were met.
- **V3 (Current Intelligent Stage):** Real-time automated IoT Replenishment. Espresso machines track exact shots pulled. Inventory drains instantly on cloud systems, automatically triggering shipments from regional depots without requiring manager validation.

4. Netflix (Entertainment Ecosystem Giant)

Process 1: Content Delivery & Streaming System

- **V1 (The Baseline):** Physical DVD-by-Mail business model. Processes were physical logistics: sorting packages, licking stamps, shipping discs through real mail.
- **V2 (First Optimization):** Centralized cloud storage streaming. Migrated data to remote servers, streaming generic media files over standard web connections to consumer devices.
- **V3 (Current Intelligent Stage): Open Connect Content Delivery Networks (CDNs).** Netflix predicts exactly what shows an area will watch tonight, copies those files to local ISP boxes during low-traffic hours, allowing high-speed localized buffering.

Process 2: Customer Retention & Discovery Process

- **V1 (The Baseline):** Simple Genre Filters (Action, Comedy, Romance). Users had to scroll alphabetically to find media titles.

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- **V2 (First Optimization):** Star-rating Collaborative Filtering. Software grouped users by shared watch history and calculated standard five-star rating vectors to predict preference.
- **V3 (Current Intelligent Stage): Hyper-Personalized Artwork Artwork System.** The actual thumbnail cover image changes per profile! If you like romance, you see a couple; if you like action, you see an explosion picture for the exact same movie.

5. Zara - Inditex (Fashion & Retail Apparel Giant)

Process 1: Design-to-Shelf Production Pipeline

- **V1 (The Baseline):** Seasonal forecasting. Fashion designs were planned 8–12 months in advance, manufactured in huge bulk batches overseas, and pushed out slowly to retail stores.
- **V2 (First Optimization):** Agile Prototyping. Zara kept raw fabrics undyed in regional centers, rapidly printing clothes in smaller batches based on early feedback cards from managers.
- **V3 (Current Intelligent Stage): Real-Time Data Pull System.** Handheld terminals used by store floor workers record customer rejections or requested sizes. Designs are modified digitally and hit global shelves in less than 15 days using automated high-speed sorting facilities.

Process 2: Retail Inventory Management

- **V1 (The Baseline):** Manual tag counting at the end of every week to check what clothes sold.
- **V2 (First Optimization):** Standard point-of-sale barcodes. Scanned pieces updated basic inventory logs at store closing.
- **V3 (Current Intelligent Stage): RFID (Radio Frequency Identification) Tracking.** Every single piece of clothing has an active tracking chip. Stores don't need to do manual inventory; automated smart scanners read the entire store's inventory room instantly inside crates.

Process Evaluation Summary Matrix

Company	Core Process Analyzed	V1 Bottleneck	V2 Optimization	V3 Ultimate Value (BPR Peak)
McDonald's	Order Fulfillment	Human Miscommunication	Electronic KDS Systems	Predictive AI Drive-Thru Planning
Amazon	Warehouse Sorting	Heavy Employee Fatigue	Kiva Robot Moving Shelves	Computer-Vision Automation
Starbucks	Customer Order Queue	High Order Discrepancies	POS Sticker System	Deep Brew AI Omni-Channel

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Company	Core Process Analyzed	V1 Bottleneck	V2 Optimization	V3 Ultimate Value (BPR Peak)
Netflix	Content Delivery	High Mail Infrastructure Cost	Basic Cloud Streams	Local Edge CDN Predictive Storage
Zara	Trend Production	Overstock / Inventory Waste	Small Batch Agility	15-Day Data Pull Fast Fashion Loops